MAE 1001

Introduction to Mechanical and Aerospace Engineering

Course website: https://gwu-mae1001.github.io/fall2026/

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GW School of Engineering & Applied Science
THE GEORGE WASHINGTON UNIVERSITY

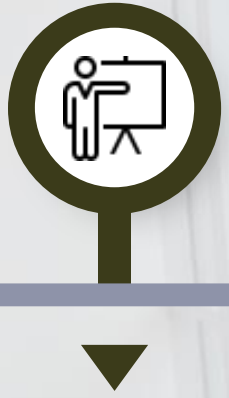
Fall 2026

Photo: Kartik Bulusu

Road Map for Class 1

30-Minute Session Blocks Breakdown

Photo Credits: Kartik Bulusu
Icon source: canva.com

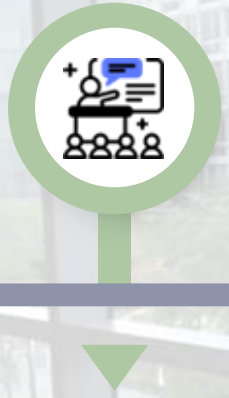


Introduction to Week 1

Announcements, overviews and more!



Blitz Talk with Will Carr



Guest Lecture with Sumner Gubsich, PhD Candidate

3D Printing and Applications



Solidworks Crash Course 1

Parts, Drawings and customizations



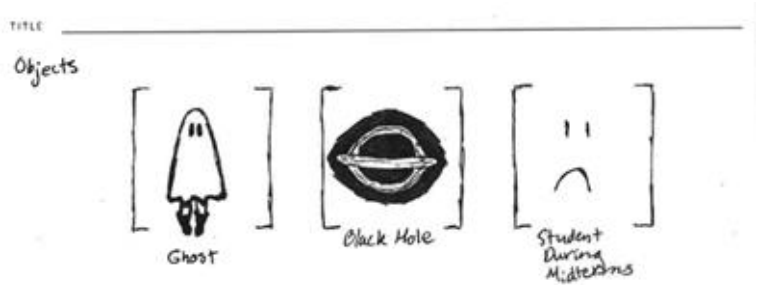
Python 101

Learn the basics of Python with Variables, Functions and Loops

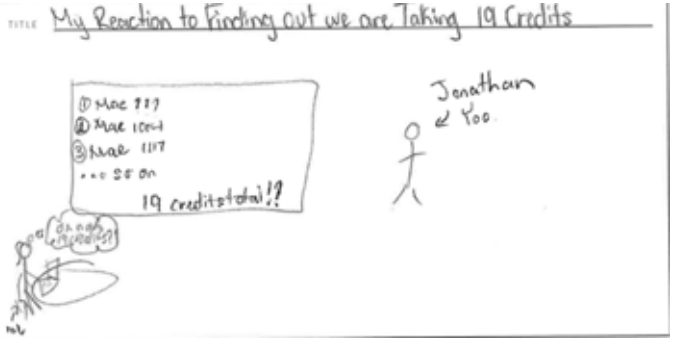
Week 1 - Best Doodles



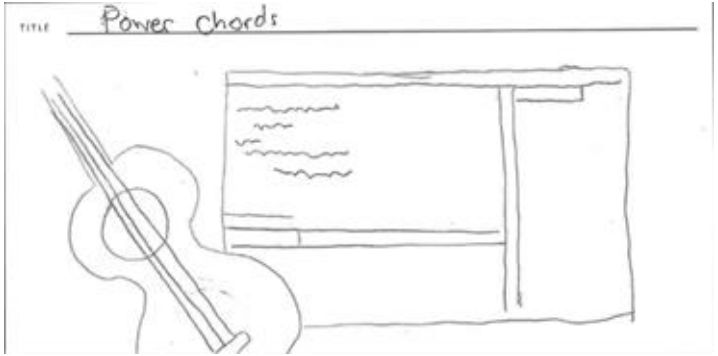
Wasim Daghmash



Gracelyn Humphreys



Uchechukwu Okeke



Lia Cruz Morillo

Guest speaker: Will Carr



Junior

Department: Mechanical and Aerospace Engineering

Email: will.carr@gwu.edu

LinkedIn: linkedin.com/in/will--carr/

I'm Will Carr, Junior in Mechanical Engineering from Northern California and love to surf and ride motorcycles at home. Here at GW I am involved in multiple Student orgs, GW Baja where I teach new members how to design and manufacture an off road race car, and GW TRAiLS where I lead outdoor trips with everything from whitewater rafting, backpacking, day hikes, and more! I also am currently working on research with SEAS and SMHS developing the PCAD medical device project.

GEORGE WASHINGTON UNIVERSITY • SEAS

CLASS OF 2028

Will Carr

Mechanical Engineering Student • GW Baja • Medical Device Research • GW TRAiLS



MOTORSPORTS
GW Baja

LAB RESEARCH
PCAD Research

LEADERSHIP
TRAILS Outdoor Trips

MOTORSPORTS ENGINEERING

GW Baja: Design, Build, Race



End-to-End Vehicle Engineering Process

CAD modeling, Manufacturing, and competitions

Hands-On Fabrication Skills

Manufacturing in the machine shop, Welding, Highbay

Competition & Industry Network

SAE intercollegiate events and automotive engineering partnerships.

LABORATORY RESEARCH

PCAD Research: Device Prototyping



Cross-Discipline Collaboration

Working with the Emergency Medicine Department

Firsthand Hardware Prototyping

Physical assembly, spring-loaded actuator tuning, and bench tests.

Academic & Industry Symposiums

Formal poster presentations to faculty and industry researchers.



OUTDOOR LEADERSHIP

TRAILS: Wilderness Expeditions



Escape the City

Get out into the nature in the DMV area

Leadership

Work with other guides to plan and lead these trips

Community

Meet outdoorsy people!





Instagram

@GWBAJA

Guest speaker: Sumner Gubisch



PhD Student

Department: Mechanical and Aerospace Engineering

Email: sumner.gubisch@gwu.edu

LinkedIn: linkedin.com/in/sumner-gubisch/

Sumner is a Ph.D. student at The George Washington University, with dual degrees from The University of Massachusetts Amherst (BS in Physics, BS in Astronomy). Currently conducting experimental and computational research on the derivation of the process-structure-property relationships of additively manufactured (laser powder bed fusion and material extrusion) semiconducting thermoelectric materials and high entropy alloys.

Week 1 - Deliverables

Week 0 10/05 to 10/11 — 10/09

Deliverables

- **Complete all installations on 10/09** and enter into a raffle
- **Week-0 review quiz on Blackboard**

Week 0 10/12 to 10/16 — 10/16

Today

Deliverables

Due in 6 days

- Python basics assignment submission on Blackboard: Download the HW1 notebook
- SolidWorks part submission on Blackboard